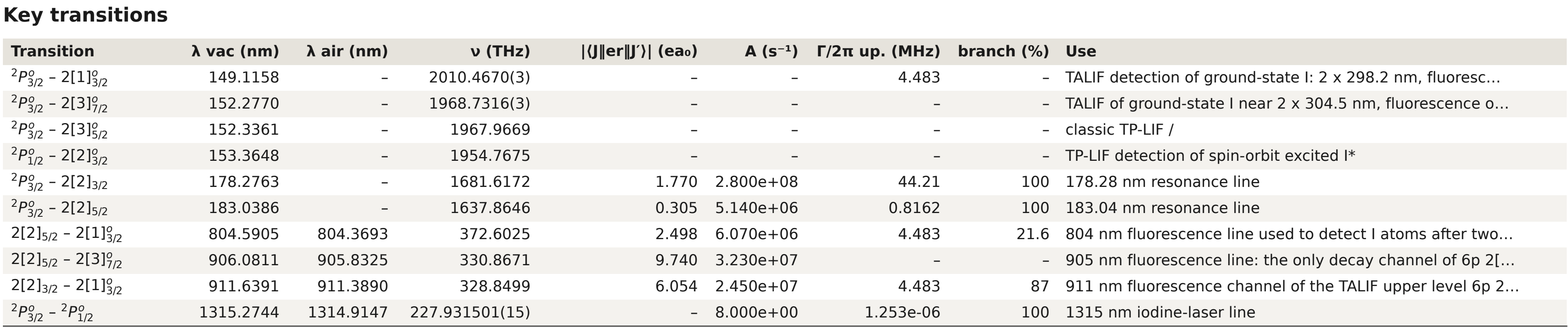
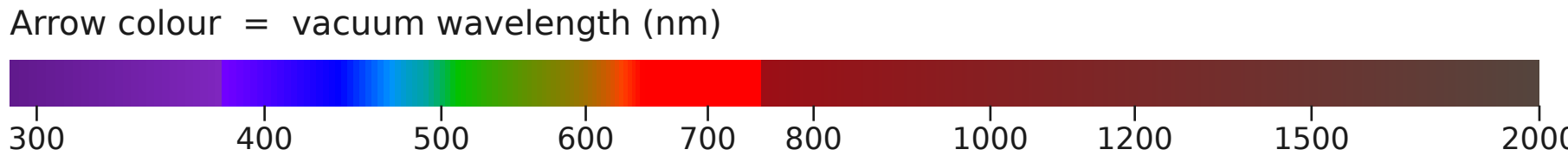


I (II). Strongest classified transitions below 2 μm (NIST): 29 levels, 50 lines.

$$\text{I}^+ \text{ ionisation limit } 84294.90 \text{ cm}^{-1} = 10.45124 \text{ eV (118.631 nm)}$$

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || e r || J' \rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

^{127}I hyperfine structure ($I = 5/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$2P_{3/2}$	827.265(3)	1146.356(10)	F=1: -3540.69	F=2: -2803.25	F=3: -837.31	F=4: +3388.83
$2P_{1/2}$	6587(4)	-	F=2: -11527.25	F=3: +8233.75		
$2[2]_{5/2}$	1417.100	-1065.500	F=0: -13145.48	F=1: -11472.66	F=2: -8222.91	F=3: -3588.03
					F=4: +2144.30	F=5: +8590.50
$2[3]_{7/2}^o$	586.700	-1104.400	F=1: -7192.02	F=2: -5703.07	F=3: -3587.99	F=4: -988.75
					F=5: +1905.30	F=6: +4857.52

Measured A, B constants; sources in the data file.